

Hiroyasu Tsukamoto

Department of Aerospace Engineering
University of Illinois Urbana-Champaign

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EDUCATION

California Institute of Technology Ph.D., Space Engineering (Autonomous Control & Robotics) with <i>Best Dissertation Award</i>	Jun. 2023
California Institute of Technology M.S., Space Engineering (Autonomous Control & Robotics)	Jun. 2018
Kyoto University B.S., Aerospace Engineering	Mar. 2017

PROFESSIONAL EXPERIENCE

Assistant Professor Aerospace Engineering, University of Illinois Urbana-Champaign	Aug. 2024 – Current
Adjunct Assistant Professor Aerospace Engineering, University of Illinois Urbana-Champaign	Sep. 2023 – Aug. 2024
Research Affiliate Maritime and Multi-Agent Autonomy, NASA Jet Propulsion Laboratory	Sep. 2023 – Aug. 2024

BIOGRAPHY

Dr. Tsukamoto is an Assistant Professor of Aerospace Engineering at the University of Illinois Urbana-Champaign and the director of the N-ACXIS Laboratory (Nonlinear Autonomous Control, Exploration, Intelligence, and Systems). Prior to joining Illinois, he was a Postdoctoral Research Affiliate in Robotics at the NASA Jet Propulsion Laboratory, where he contributed to the Science-Infused Spacecraft Autonomy for Interstellar Object Exploration and Multi-Spacecraft Autonomy Technology Demonstration projects. He received his Ph.D. and M.S. in Space Engineering, advised by Prof. Soon-Jo Chung. He is the recipient of several awards, including the **William F. Ballhaus Prize for the Outstanding Doctoral Dissertation in Space Engineering at Caltech**, Forbes 30 Under 30 Asia – Class of 2025, the Innovators Under 35 Japan Award from MIT Technology Review, and the AIAA Journal of Guidance, Control, and Dynamics Editor's Choice Award. He is an Associate Editor of the IEEE Robotics and Automation Letters.

SELECTED RESEARCH GRANTS

Hybrid Human-Machine Autonomy for Resilient Urban Mobility and Sustainability

- Role: Co-PI
- Source: Brain Pool, National Research Foundation of Korea
- Amount: 1 postdoctoral salary for 3 years and 6 months of Co-PI salary per year for 3 years.
- Period of Performance: 3 years

SCRIPTED-FM: Selective and Contractive Robotic Intelligence for Performant, Trustworthy, and Exploratory Decisions with Foundation Models

- Role: PI

- Source: Safe and Assured Foundation Robots for Open Environments (SAFRON), DARPA
- Amount: \$113,479.00
- Period of Performance: 1 year

SELECTED PUBLICATIONS (<https://scholar.google.com/citations?user=G9iATfcAAAAJ>)

Peer-Reviewed Journal Articles

1. T.-W. Hsu and **H. Tsukamoto**, “Statistical guarantees in data-driven nonlinear control: Conformal robustness for stability and safety,” *IEEE Control System Letters*, Mar. 2025.
 - DOI: [10.1109/LCSYS.2025.3578062](https://doi.org/10.1109/LCSYS.2025.3578062)
 - Peer-reviewed and accepted also at the *IEEE Conference on Decision and Control*
2. **H. Tsukamoto**, S.-J. Chung, B. Donitz, M. Ingham, D. Mages, and Y. K. Nakka, “Neural-Rendezvous: Learning-based robust guidance and control to encounter interstellar objects,” *AIAA Journal of Guidance, Control, and Dynamics*, Oct. 2024.
 - DOI: [10.2514/1.G007671](https://doi.org/10.2514/1.G007671)
 - ***AIAA Journal of Guidance, Control, and Dynamics Editor’s Choice Award***
3. **H. Tsukamoto**, S.-J. Chung, and J.-J. E. Slotine, “Contraction theory for nonlinear stability analysis and learning-based control: A tutorial overview,” *Annual Reviews in Control*, Nov. 2021.
 - DOI: [10.1016/j.arcontrol.2021.10.001](https://doi.org/10.1016/j.arcontrol.2021.10.001)
 - ***Tutorial paper invited by the journal’s Editor-in-Chief***
4. **H. Tsukamoto**, S.-J. Chung and J.-J. E. Slotine, “Neural stochastic contraction metrics for learning-based control and estimation,” *IEEE Control Systems Letters*, Nov. 2021.
 - DOI: [10.1109/LCSYS.2020.3046529](https://doi.org/10.1109/LCSYS.2020.3046529)
 - Peer-reviewed and accepted also at the *American Control Conference*
5. **H. Tsukamoto** and S.-J. Chung, “Learning-based robust motion planning with guaranteed stability: A contraction theory approach,” *IEEE Robotics and Automation Letters*, Oct. 2021.
 - DOI: [10.1109/LRA.2021.3091019](https://doi.org/10.1109/LRA.2021.3091019)
 - Peer-reviewed and accepted also at the *IEEE International Conference on Intelligent Robots and Systems*
6. **H. Tsukamoto** and S.-J. Chung, “Robust controller design for stochastic nonlinear systems via convex optimization,” *IEEE Transactions on Automatic Control*, Oct. 2021.
 - DOI: [10.1109/TAC.2020.3038402](https://doi.org/10.1109/TAC.2020.3038402)
7. **H. Tsukamoto** and S.-J. Chung, “Neural contraction metrics for robust estimation and control: A convex optimization approach,” *IEEE Control Systems Letters*, Jan. 2021.
 - DOI: [10.1109/LCSYS.2020.3001646](https://doi.org/10.1109/LCSYS.2020.3001646)
 - Peer-reviewed and accepted also at the *IEEE Conference on Decision and Control*

Peer-Reviewed Conference Papers

8. R. D’Silva and **H. Tsukamoto**, “Statistical contraction for chance-constrained trajectory optimization of non-Gaussian stochastic systems,” *accepted, IEEE International Conference on Robotics and Automation*, Jun. 2026.
 - Preprint: [arXiv:2603.07092](https://arxiv.org/abs/2603.07092)

9. K. Hirano and **H. Tsukamoto**, “Conformal Koopman for embedded nonlinear control with statistical robustness: Theory and real-world validation,” *accepted, IEEE International Conference on Robotics and Automation*, Jun. 2026.
 - Preprint: [Google Drive](#)
10. S. Wei, M. Ornik, and **H. Tsukamoto**, “Conformal contraction for robust nonlinear control with distribution-free uncertainty quantification,” *accepted, IEEE Conference on Decision and Control*, Dec. 2025.
 - DOI: [10.1109/CDC57313.2025.11312813](#)
11. A. Bhardwaj, S. Bhatta, and **H. Tsukamoto**, “Information-optimal multi-spacecraft positioning for interstellar object exploration,” *IEEE Aerospace Conference*, Mar. 2025.
 - DOI: [10.1109/AERO63441.2025.11068630](#)
12. **H. Tsukamoto***, J. Hajar*, S.-J. Chung, and F. Y. Hadaegh (*co-first authors), “Regret-optimal defense against stealthy adversaries: A system level approach,” *IEEE Conference on Decision and Control*, 2023.
 - DOI: [10.1109/CDC56724.2024.10886620](#)
13. **H. Tsukamoto**, J. D. Ibrahim, J. Hajar, J. Ragan, S.-J. Chung, and F. Y. Hadaegh, “Robust optimal network switching topology for zero dynamics attacks,” *IEEE Conference on Decision and Control*, Dec. 2023.
 - DOI: [10.1109/CDC56724.2024.10886460](#)
14. **H. Tsukamoto**, B. Riviere, C. Choi, A. Rahmani, and S.-J. Chung, “CaRT: Certified safety and robust tracking in learning-based motion planning for multi-agent systems,” *IEEE Conference on Decision and Control*, Dec. 2023.
 - DOI: [10.1109/CDC49753.2023.10383600](#)
15. B. Donitz, D. Mages, **H. Tsukamoto**, P. Dixon, D. Landau, S.-J. Chung, E. Bufanda, M. Ingham, J. Castillo-Rogez, “Interstellar object accessibility and mission design,” *IEEE Aerospace Conference*, Mar. 2023.
 - DOI: [10.1109/AERO55745.2023.10115554](#)
16. **H. Tsukamoto**, S.-J. Chung, J.-J. E. Slotine, and C. Fan, “A theoretical overview of neural contraction metrics for learning-based control with guaranteed stability,” *IEEE Conference on Decision and Control*, Dec. 2021.
 - DOI: [10.1109/CDC45484.2021.9682859](#)
 - Featured at the IEEE CDC tutorial session ([Contraction Theory for Machine Learning](#))
17. S. Singh, **H. Tsukamoto**, B. Lopez, S.-J. Chung, and J.-J. E. Slotine, “Safe motion planning with tubes and contraction metrics,” *IEEE Conference on Decision and Control*, Dec. 2021.
 - DOI: [10.1109/CDC45484.2021.9682865](#)
 - Featured at the IEEE CDC tutorial session ([Contraction Theory for Machine Learning](#))
18. **H. Tsukamoto**, S.-J. Chung, and J.-J. E. Slotine, “Learning-based adaptive control using contraction theory,” *IEEE Conference on Decision and Control*, Dec. 2021.
 - DOI: [10.1109/CDC45484.2021.9683435](#)
19. **H. Tsukamoto** and S.-J. Chung, “Convex optimization-based controller design for stochastic nonlinear systems using contraction analysis,” *IEEE Conference on Decision and Control*, Dec. 2019.
 - DOI: [10.1109/CDC40024.2019.9028942](#)

Selected Papers under Review

20. M. Cho, **H. Tsukamoto**, and H. T. Tran, “CARL: Contraction-aware reinforcement learning for robust path-tracking,” *IEEE Transactions on Robotics*.
21. M. Vora, G. Puthumanaillam, **H. Tsukamoto**, M. Ornik, “SCoUT: Scalable communication via utility-guided temporal grouping in multi-agent reinforcement learning,” *International Conference on Machine Learning*.

SELECTED HONORS/AWARDS

AIAA Journal of Guidance, Control, and Dynamics Editor’s Choice Award	Jul. 2025
• Selected by the journal’s Editors-in-Chief as one of the outstanding papers of the quarter	
Forbes 30 Under 30 Asia – Class of 2025	Mar. 2025
• Named one of Asia-Pacific’s top 30 in Healthcare & Science <u>under age 30</u> (4500+ nominees)	
Dentsu Soken Outstanding Innovator Award for IU35 MIT Technology Review	Dec. 2024
• Best innovator award among the Innovators Under 35 Japan awardees	
Innovators Under 35 Japan Award from MIT Technology Review	Nov. 2024
• Named one of Japan’s top 10 innovators <u>under age 35</u> (200+ nominees)	
Best Tutorial Award at IEEE Systems, Man, and Cybernetics (SMC)	Mar. 2024
• Best tutorial award for “ML-based Control for Autonomous and Robotic Systems”	
William F. Ballhaus Outstanding Dissertation Prize	Jun. 2023
• Best doctoral dissertation in Space Engineering, GALCIT, Caltech	
IEEE Conference on Decision and Control (CDC) Student Travel Support Award	Dec. 2021
• Full financial support for IEEE CDC travel expenses and registration fees	
IEEE American Control Conference (ACC) Student Travel Support Award	May 2021
• Full financial support for IEEE ACC travel expenses and registration fees	
IEEE Conference on Decision and Control Student Travel Support Award	Dec. 2020
• Full financial support for IEEE CDC travel expenses and registration fees	
Honjo International Scholarship Award (2% acceptance rate)	Aug. 2020
• Financial support for graduate living expenses (\$48,000)	
Caltech Vought Fellowship	Oct. 2017
• Financial support for graduate living expenses (\$23,800) and tuition (\$48,111)	
Funai Overseas Scholarship Award (9% acceptance rate)	Sep. 2017
• Financial support for graduate living expenses (\$60,000) and tuition (about \$100,000)	
Kanae Scholarship Award from Kyoto University President	Oct. 2016
• Full financial support for visiting US top 10 aerospace graduate programs	
HeKKSaGOn Scholarship Award from Kyoto University President	Feb. 2016
• Full financial support for research internship at University of Heidelberg	
Kyoto University John Mung Scholarship Award	Aug. 2015
• Full financial support for summer school program at University of Oxford	
Kyoto University John Mung Scholarship Award	Mar. 2015
• Full financial support for spring school program at Stanford University	